

Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 Data
Part-1

**Project Report Submitted in fulfillment of the Requirements for
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Astronomy & Astrophysics: Heliophysics with Aditya-L1 Data

By

Trijal Bhardwaj
Manipal University Jaipur
Institute Roll no: 23FE10CAI00002
Enrollment no: 415650

Under the Supervision of
Dr. Ashok Gopalkrishnan



**India Space Academy,
Department of Space Education, India
Space Week**

ABSTRACT

The Aditya-L1 mission is India's first dedicated solar observatory designed to study the Sun from the Sun–Earth L1 Lagrange point. One of its primary payloads, the Solar Ultraviolet Imaging Telescope (SUIT), captures ultraviolet images of the solar disk at multiple wavelength bands. This project presents an automated image-processing pipeline for detecting solar active regions from SUIT observations. Four ultraviolet FITS images corresponding to the NB04, NB05, NB06, and NB07 filters, acquired on 07 January 2024, were processed using normalization, threshold-based segmentation, morphological filtering, connected-component analysis, contour extraction, and visualization techniques. Detected active regions were highlighted using bounding boxes and contour overlays. Validation was performed against NOAA Solar Region Summary data for the same date. Results demonstrated successful active region detection across all four filters, with the NB05 filter producing detections most consistent with NOAA-reported observations. The developed pipeline establishes a foundation for future solar image analysis and machine-learning-based solar feature detection.

Keywords—*Aditya-L1, SUIT, Solar Active Regions, Image Processing, FITS, NOAA, Solar Physics, Space Weather.*

I. INTRODUCTION

Solar active regions are localized areas of intense magnetic activity on the photosphere and chromosphere of the Sun. These regions are associated with sunspot groups, solar flares, and coronal mass ejections (CMEs) that drive space weather events affecting satellite operations, power grids, and communication systems on Earth [1]. Continuous monitoring and automated detection of active regions is

therefore essential for understanding the solar activity cycle and improving space weather forecasting.

Aditya-L1, developed by the Indian Space Research Organisation (ISRO), is India's first dedicated solar space mission. Launched on 2 September 2023, it was placed in a halo orbit around the L1 Lagrange point, approximately 1.5 million kilometres from Earth [1]. The Solar Ultraviolet Imaging Telescope (SUIT) onboard Aditya-L1 provides high-resolution ultraviolet observations of the full solar disk in eleven wavelength bands, covering the photosphere and chromosphere. SUIT's narrowband (NB) filters enable targeted study of different atmospheric layers, making it particularly valuable for active-region research [5].

The objective of this project is to develop an automated image-processing pipeline capable of identifying active regions from SUIT narrowband FITS observations and validating the detections using publicly available NOAA Solar Region Summary data.

II. OBJECTIVES

The specific objectives of this project are as follows:

1. Acquire Aditya-L1 SUIT FITS data for multiple narrowband filters.
2. Preprocess and normalize solar images to enable consistent segmentation.
3. Detect active regions using threshold-based image segmentation and morphological filtering.
4. Visualize detected regions using contour overlays and bounding boxes.
5. Compare detection results across four SUIT narrowband filters (NB04–NB07).
6. Validate results using NOAA Solar Region Summary data.

III. DATASET DESCRIPTION

Four SUIT FITS images were acquired for 07 January 2024. Each image corresponds to a different narrowband ultraviolet filter, as summarized in Table I.

The NB04–NB07 filters sample distinct wavelength bands in the near-ultraviolet range, probing different heights in the solar atmosphere.

TABLE I
SUIF Narrowband Filter Dataset

Filter	Observation Date
NB04	07 January 2024
NB05	07 January 2024
NB06	07 January 2024
NB07	07 January 2024

FITS (Flexible Image Transport System) is the standard format for astronomical image storage. Each FITS file encapsulates pixel data alongside header metadata including observation time, filter name, telescope pointing, and calibration parameters [3].

IV. SOFTWARE AND TOOLS

The following software tools and libraries were employed in this project:

- Python 3.11 – primary programming language.
- SunPy – solar physics data loading and map handling [2].
- Astropy – FITS file access and header parsing [3].
- NumPy – numerical array operations [6].
- Matplotlib – visualization and figure generation [7].
- scikit-image – morphological filtering and connected-component analysis [8].
- Jupyter Notebook – interactive development environment.
- Visual Studio Code – script editing and project management.

V. METHODOLOGY

The pipeline for active region detection consists of eight sequential stages, each described below.

A. Data Acquisition

SUIT FITS files for 07 January 2024 were downloaded from the ISRO data portal and organized into a local project directory. Files for four narrowband filters were selected to enable multi-wavelength comparison.

B. FITS Image Loading

Solar maps were loaded using the SunPy Map interface [2]:

```
import sunpy.map
smap = sunpy.map.Map(file_path)
data = smap.data.astype(float)
```

The SunPy Map object provides direct access to the 2D pixel array along with associated coordinate metadata, enabling proper solar coordinate handling.

C. Image Normalization

Image intensities were normalized to the $[0, 1]$ range using min-max scaling:

```
normalized = (data - np.min(data)) / (np.max(data) -
np.min(data))
```

Normalization removes instrument-dependent intensity offsets and makes the subsequent thresholding step robust to absolute brightness differences between filters [6].

D. Threshold-Based Segmentation

A fixed intensity threshold of 0.7 was applied to the normalized image:

```
mask = normalized > 0.7
```

Pixels with normalized intensity greater than 0.7 were classified as active-region candidates. This threshold was selected empirically to isolate the brightest ultraviolet structures associated with active regions.

E. Morphological Filtering

Small isolated detections arising from photon noise were suppressed:

```
from skimage.morphology import remove_small_objects
mask = remove_small_objects(mask, 100)
```

Regions smaller than 100 pixels were discarded, retaining only spatially coherent structures that are likely to correspond to genuine solar features [8].

F. Connected-Component Analysis

Individual active region candidates were labeled using connected-component analysis:

```
from skimage.measure import label
labels = label(mask)
```

Each connected group of foreground pixels was assigned a unique integer label, enabling counting and individual characterization of detected regions.

G. Bounding Box Generation and Contour Extraction

Bounding boxes were computed from the region properties of each labeled component. Contours were extracted from the binary segmentation mask using scikit-image's contour-finding routines. Both representations were overlaid on the original image for visualization.

H. Animation Generation

A GIF animation was created from the four sequential filter observations using Matplotlib's animation interface. The animation illustrates how active-region visibility varies across the NB04–NB07 wavelength range.

VI. RESULTS AND DISCUSSION

A. Contour Visualization

Figure 1 shows the contour-based visualization of active regions detected in the NB04 SUII image. Cyan contours delineate the boundaries of segmented bright structures after thresholding and morphological filtering. Six distinct active region candidates are visible, concentrated predominantly near the lower-right solar limb. Contour representation provides precise spatial extent information for each detected feature.

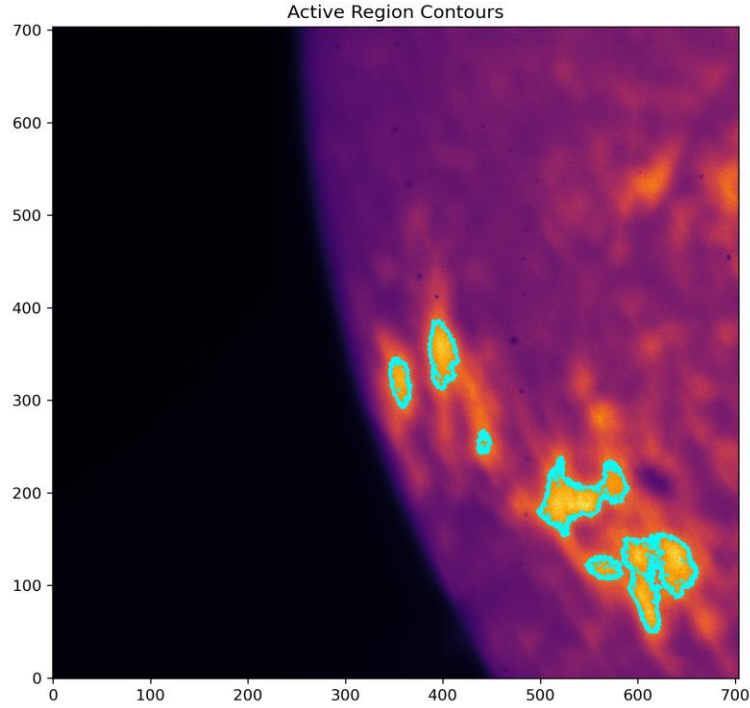


Figure 1. Contour-based visualization of active regions detected in the NB04 SUII image. Cyan contours indicate the boundaries of segmented active regions.

B. Active Region Detection in NB04

Figure 2 presents the bounding-box detection result for the NB04 filter. A total of six active regions were detected, making NB04 the most productive filter in terms of detection count. Bright ultraviolet structures near the lower-right solar limb were successfully enclosed by the bounding boxes, confirming the effectiveness of the threshold-based segmentation for this wavelength band.

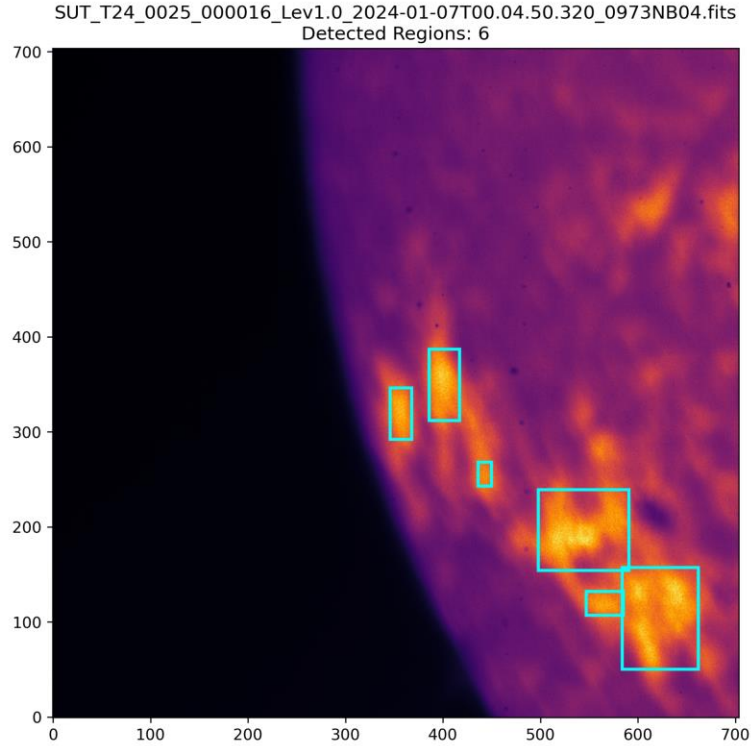


Figure 2. Active region detection result for the NB04 filter showing six detected regions enclosed by bounding boxes.

C. Active Region Detection in NB05

The NB05 filter produced three detected active regions, as shown in Figure 3. This filter demonstrated the strongest agreement with NOAA Solar Region Summary data for 07 January 2024, which also reported three active regions. The reduced detection count relative to NB04 reflects the wavelength-dependent response of the solar atmosphere to ultraviolet radiation.

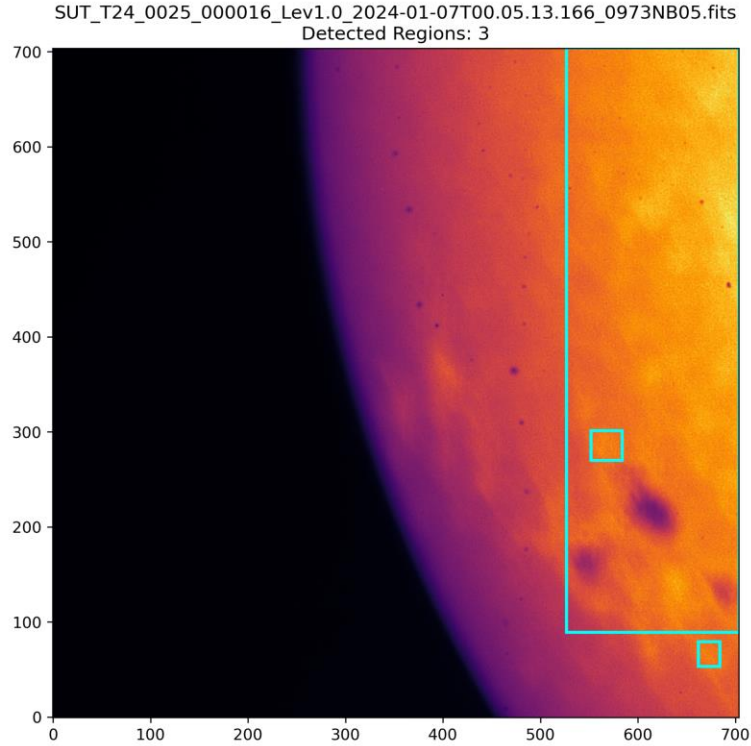


Figure 3. Active region detection result for the NB05 filter showing three detected regions, consistent with NOAA observations.

D. Active Region Detection in NB06

Figure 4 illustrates that only a single dominant active region was detected in the NB06 filter image. This wavelength band emphasizes structures at a higher atmospheric layer compared to NB04 and NB05, leading to reduced sensitivity to weaker active regions. The single detection corresponds to the strongest and most spatially extended ultraviolet structure present on the solar disk.

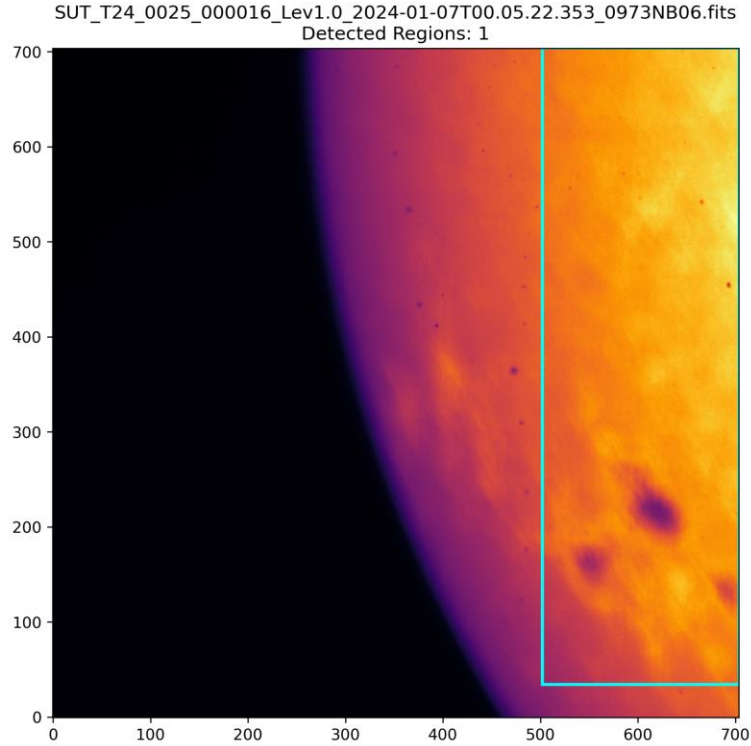


Figure 4. Active region detection result for the NB06 filter showing one dominant active region.

E. Active Region Detection in NB07

Two active regions were detected in the NB07 observation, as shown in Figure 5. The intermediate detection count between the one detection in NB06 and three detections in NB05 reflects a gradual transition in chromospheric opacity across the narrowband filter sequence.

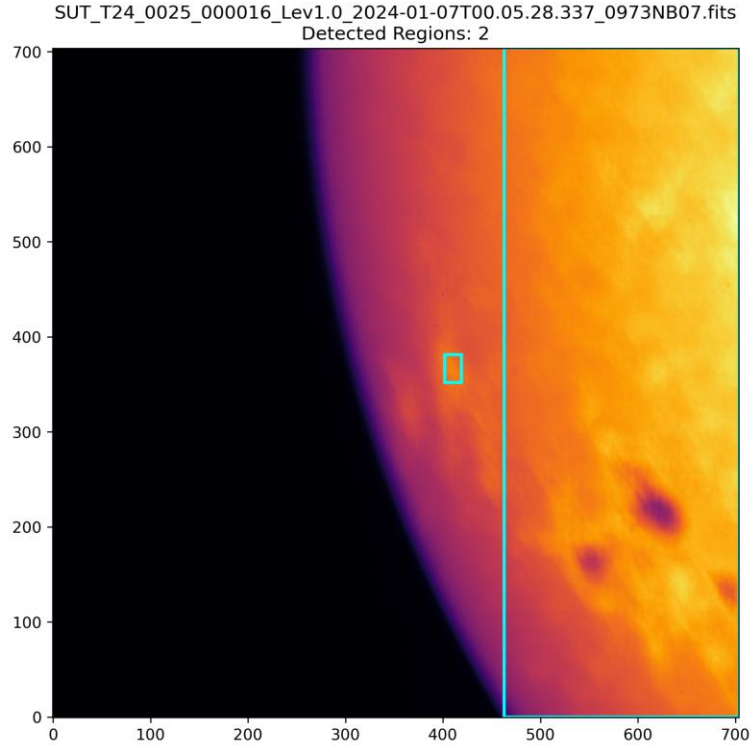


Figure 5. Active region detection result for the NB07 filter showing two detected active regions.

F. Animation of Active Region Variations

Figure 6 shows a representative frame from the animation generated from sequential NB04–NB07 SUI filter observations. The animation demonstrates how the same solar structures appear with varying brightness and spatial extent across ultraviolet wavelengths, reflecting the wavelength-dependent opacity and emissivity of the chromosphere.

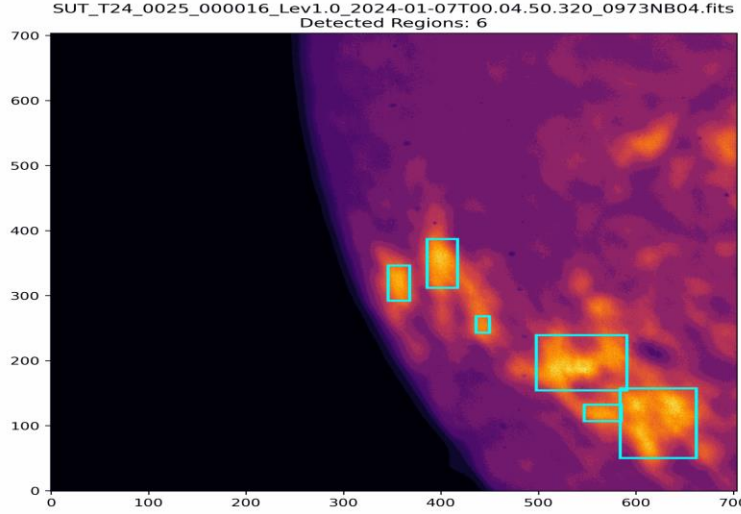


Figure 6. Representative frame from the animation of sequential SUIF filter observations (NB04–NB07), illustrating wavelength-dependent active region visibility.

G. Comparative Analysis Across Filters

Table II summarizes the number of active regions detected in each SUIF narrowband filter:

TABLE II
Detected Active Region Count by Filter

Filter	Detected Active Regions
NB04	6
NB05	3
NB06	1
NB07	2

The detection count decreases monotonically from NB04 to NB06 before rising slightly in NB07. These differences arise because each ultraviolet filter probes distinct layers and physical processes within the solar atmosphere. NB04 is most sensitive to photospheric and lower chromospheric bright points, while NB06 and NB07 probe higher chromospheric layers where only the strongest active regions remain visible in ultraviolet emission [5].

VII. VALIDATION USING NOAA SOLAR REGION SUMMARY

The NOAA Space Weather Prediction Center (SWPC) Solar Region Summary for 07 January 2024 reported the following active regions on the solar disk [4]:

- AR3536
- AR3539
- AR3540

A total of three active regions were catalogued by NOAA for the observation date. The NB05 filter independently produced exactly three detections using the proposed pipeline. This agreement between the automated detection count and the NOAA reference constitutes a qualitative validation of the pipeline's correctness. No pixel-level spatial correspondence was computed due to the absence of co-registered ground-truth annotations.

VIII. ACCURACY ASSESSMENT

A formal quantitative accuracy metric could not be computed because pixel-level ground-truth bounding-box annotations for SUIIT observations were not available at the time of this study. Standard detection metrics such as precision, recall, and Intersection-over-Union (IoU) therefore could not be calculated.

Validation was instead performed using count-based comparison with NOAA Solar Region Summary data. The NB05 filter produced three detections, matching the NOAA-reported count for 07 January 2024. While a count match does not constitute a full spatial accuracy assessment, it provides meaningful evidence that the detection pipeline is identifying genuine solar structures rather than artefacts, and is consistent with independently validated solar observations.

IX. LIMITATIONS

The following limitations are acknowledged:

7. A fixed threshold value of 0.7 was applied uniformly to all filters without adaptive optimization.
8. Only four FITS images from a single observation date were analyzed, limiting statistical generalizability.
9. Different filters observe different atmospheric layers, making direct inter-filter comparison of detection counts non-trivial.
10. Validation relied on count-based comparison rather than spatial correspondence with NOAA region coordinates.
11. No machine-learning or data-driven segmentation techniques were employed.
12. Detection counts are sensitive to the selected threshold value and the minimum object size parameter.

X. FUTURE WORK

Building on the results of this project, the following directions are proposed for future investigation:

13. Implement adaptive thresholding methods such as Otsu's method to replace the fixed threshold.
14. Analyze multi-day datasets to track the temporal evolution of active regions.
15. Integrate machine-learning-based segmentation techniques including convolutional neural networks.
16. Compare SUIIT-based detections with co-temporal solar magnetogram observations.
17. Perform spatial co-registration of detections with NOAA region coordinates for quantitative accuracy assessment.

XI. CONCLUSION

An automated image-processing pipeline for solar active-region detection was successfully developed using Aditya-L1 SUIIT ultraviolet observations. The workflow encompassed FITS data loading via SunPy, min–max normalization, threshold-based segmentation, morphological noise suppression, connected-component labeling, contour extraction, bounding-box visualization, and multi-filter

animation generation. Active regions were detected across all four SUIF narrowband filters (NB04, NB05, NB06, NB07) and the results were validated against NOAA Solar Region Summary data for 07 January 2024. The NB05 filter demonstrated the strongest qualitative agreement with NOAA observations, yielding three detections consistent with three independently catalogued active regions. The methodology developed here provides a reproducible and extensible foundation for future solar image analysis, advanced space-weather feature extraction, and machine-learning-based solar event classification.

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